JANIE QING

janie.qing@duke.edu | (757) 373-9226 | https://www.linkedin.com/in/janie-qing/

EDUCATION

Duke University, Angier B. Duke Scholar (Full-Ride Merit Scholarship) **May 2028**
B.S.E in Electrical and Computer Engineering and B.S. in Computer Science Durham, NC
GPA: 3.8/4.0

- **Relevant Coursework:** Computer Architecture, Fund. Of Circuits, Signals & Systems, Data Structures, Physics, Dynamics

University of Oxford, New College Oxford, UK

- **Relevant Coursework:** The Ethics and Philosophy of Artificial Intelligence

WORK EXPERIENCE

Boston Dynamics **July 2026 – Present**
Systems Test Engineering Co-Op Waltham, MA

- Executing systems-engineering test procedures, stands, and systems for Atlas, an electrically actuated humanoid robot
- Wiring 16 thermal couplers onto a gripper-hand for thermal characterization testing and authoring a program to cycle the hand through positions while sweeping D-current to emulate torque loads and log the thermal response
- Reverse-engineering a magnetic powder-brake tension controller for programmatic control, tracing encoder/MCU signals and characterizing its MOSFET PWM drive on an oscilloscope to enable driving it from an external controller

NASA – Langley Research Center **May 2026 – July 2026**
Hardware Engineering Intern (OSTEM), Space Robotics Hampton, VA

- Rebuilt two robotic end-effectors, swapping 3D-printed designs for self-altered (drilling, tapping, filing) metal parts to improve thermal vacuum readiness from 64% to 95% with the replacement of ~12 parts for each end-effector
- Fabricated metal electronics enclosure and debugged motor-driven telescoping rods, isolating a limit-switch interference
- Defined a UR3e robotic join sequence to autonomously join six panels into a modular cubo-octehedron unit with repeatable alignment, modeling and visualizing different assembly configurations and approaches in Creo and Blender

SIG Sauer – Drone R&D **Jan 2026 – May 2026**
Robotics Engineering Intern Sanford, NC

- Designed and fabricated custom test jigs and fixtures to support repeatable electronics assembly and validation of UAVs
- Performed structured evaluation and fault isolation of electromechanical assemblies to validate power distribution subsystems, tracing faults across wiring, connectors, and power paths, preparing 9 5-inch drones for a training course
- Developed propulsion characterization protocols by building and wiring up a thrust stand to analyze efficiency, load response, and performance for 6 different configurations of motors and propellers

Lumius Imaging (YC P26) **Jan 2026 – May 2026**
Engineering Intern Durham, NC

- Prototyped leak-proof enclosure hardware for Lumius' fluid-filled, acoustically coupled 3D ultrasound probe, modeling the housing and a sliding magnetic leak-proof imaging window in Fusion 360 and preparing a prototype for YC interview

Ni Lab at Duke University **Aug 2025 – Dec 2025**
Engineering Intern Durham, NC

- Assembled and reworked sensor hardware for a flexible, skin-mounted mechano-acoustic sensor under a microscope, soldering fine-pitch components and integrating a triaxial accelerometer (1600 Hz) for continuous signal capture

PROJECTS

Autonomous Line Sensing Robot **Jan 2025 – May 2025**

- Built and programmed an Arduino-based autonomous robot achieving 98% line-tracking accuracy and ±2 mm path precision using QTI reflectance sensors, proportional motor control, and real-time calibration algorithms
- Integrated wireless communication and sensing subsystems, enabling high object classification accuracy, <80 ms Xbee data latency, and <0.3s synchronization between four other robots during multi-agent coordination tasks

Bubble CPAP Liquid Level Monitoring Device – Makerere University, Uganda **Sep 2024 – Dec 2024**

- Programmed an Arduino Nano in C++ (HX711 library) to read load-cell weight, zero scale, and fire an audible buzzer alert
- Wired and soldered full sensing circuit – load cell + HX711 amplifier, three status/alarm LED's, buzzer, tare button, power

LEADERSHIP & ACTIVITIES

Asian Student Association **Aug 2024 – May 2026**
Executive Team for Community and Outreach Durham, NC

- Organizing community-building events and programs for 300+ people to promote inclusivity and cultural awareness

Involvements: Theta Tau, 180 Degrees Consulting, Catalyst (Duke's Pre-Professional Tech Org), Society of Women Engineers

SKILLS

Technical Skills: Altium (PCB Design), Circuitry, SolidWorks, Catia, Blender, Fusion360, Soldering, Oscilloscope
Programming Languages: C++, Java, Python, MATLAB